

Static Timing Analysis

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Course Overview

1. STA Concepts
 - 1 lecture
2. Delay Modeling
 - 1 lecture
3. Interconnect Parasitics
 - 1 lecture
4. Delay Calculation
 - 1 lecture
5. Configuring the STA Environment
 - 1 lecture
6. **Timing Checks**
 - 1 lecture
7. Crosstalk and Noise
 - 1 lecture

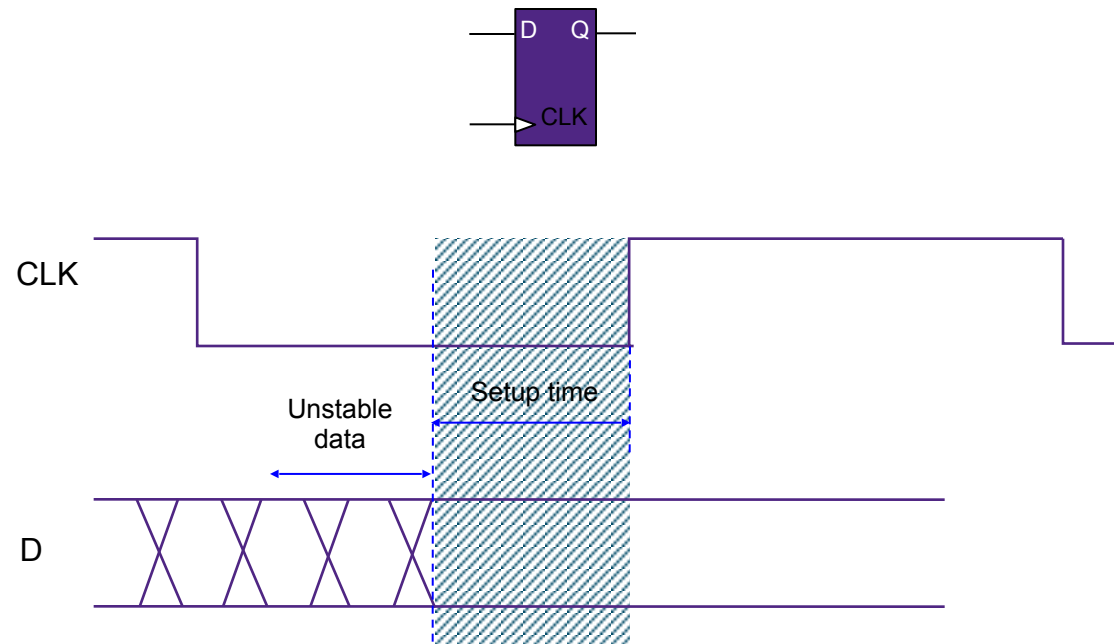
6. Timing Checks

Timing Checks

- Constraints set by sequential circuits:
 - Setup
 - Hold
- Possible conditions:
 - Best-case (Hold)
 - Worst-case (Setup)

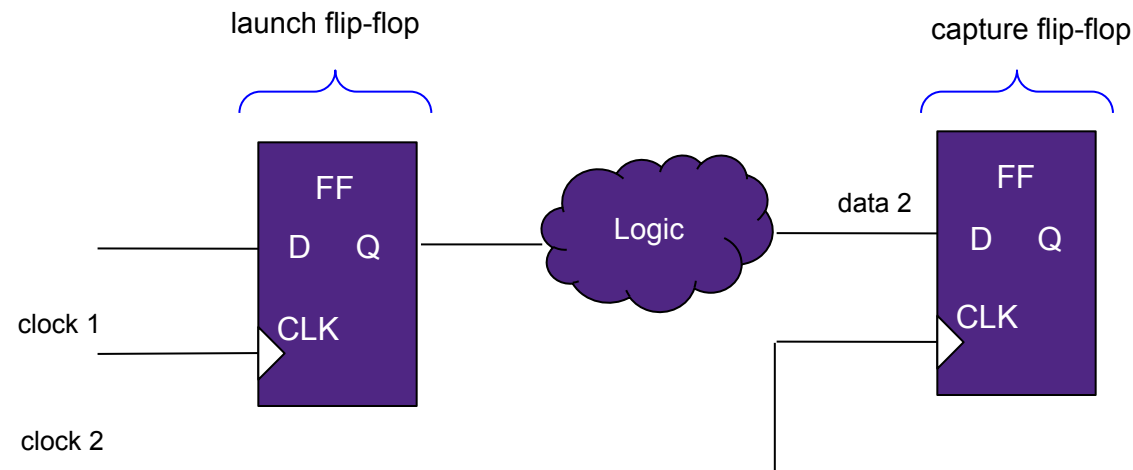
Setup Timing Check

- Data must be stable before the active edge of the clock



Launch and Capture Flip-flops

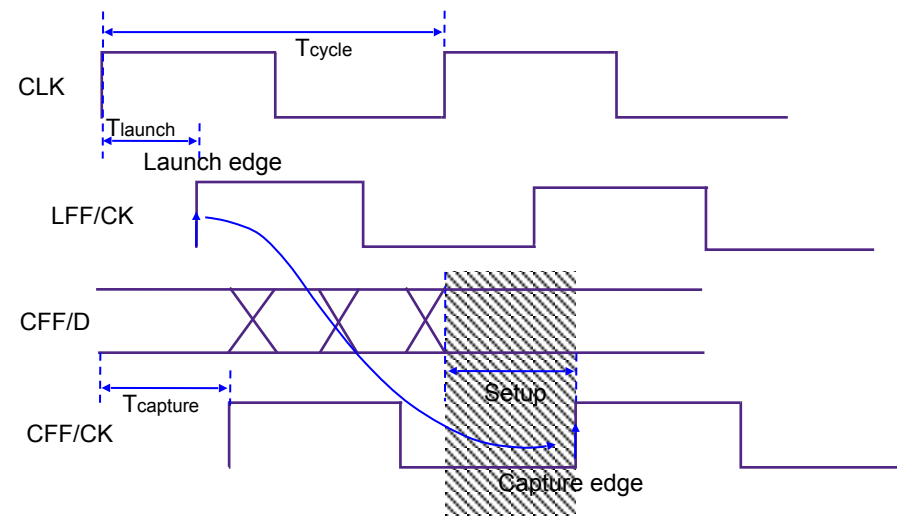
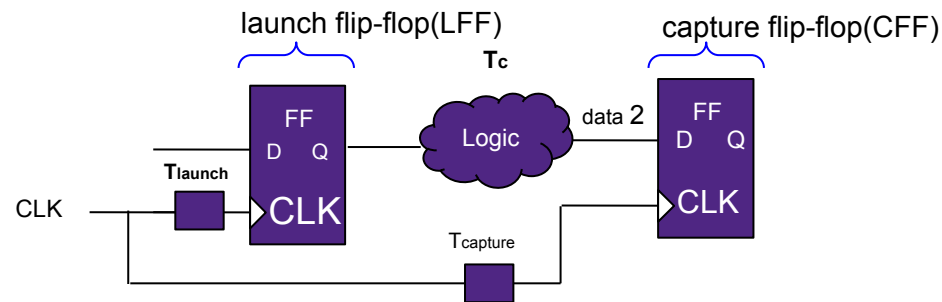
- Setup is checked from first active edge (clock) of launch flip-flop to closest active edge of capture flip-flop



Data and Clock Signals for Setup Timing Check

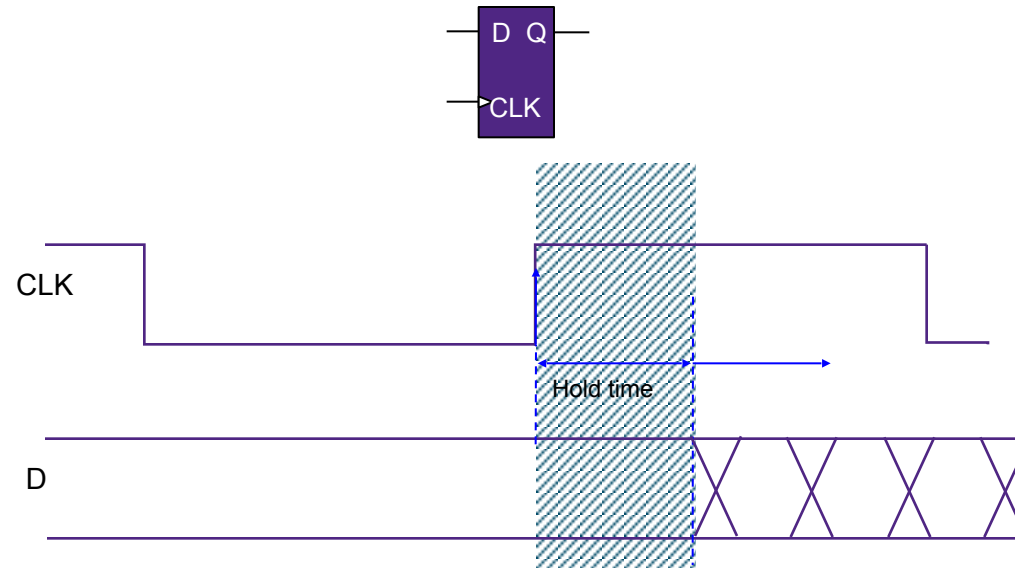
Setup condition

$$T_{LFF} + T_{\text{launch}} + T_{LFF} + T_c < T_{\text{capture}} + T_{\text{cycle}} - T_{\text{setup}}$$



Hold Timing Check

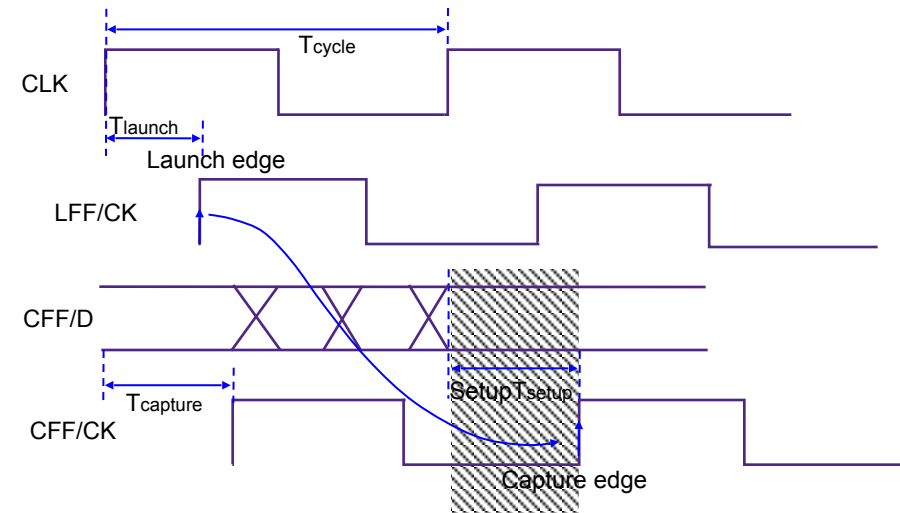
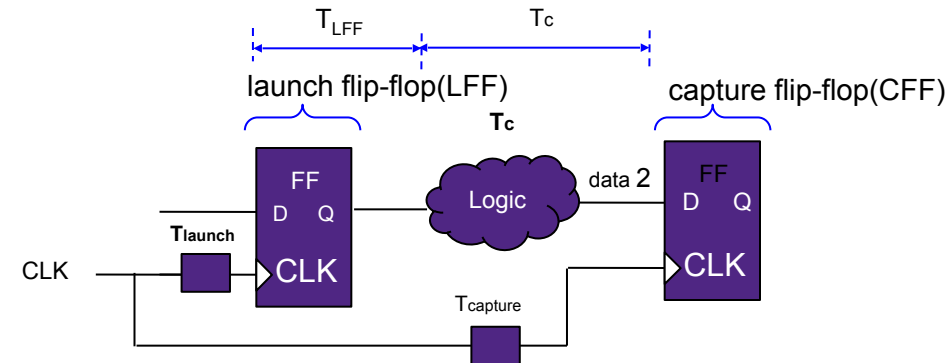
- Verifies that the data is held stable for a specified amount of time after the active edge of the clock



Data and Clock Signals for Hold Timing Check

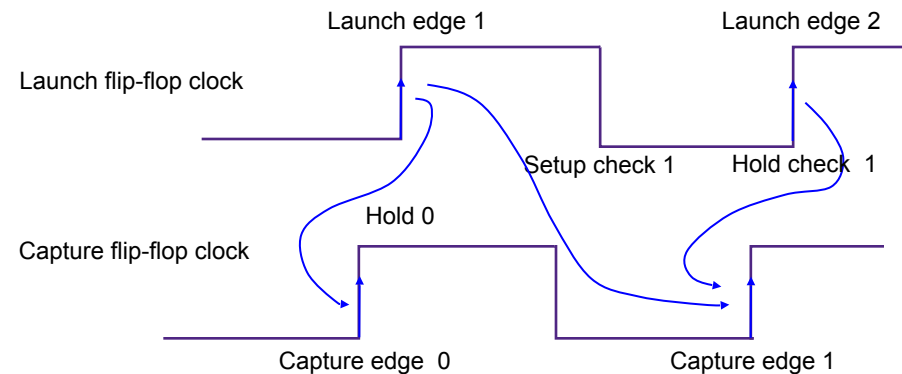
Hold condition

$$T_{\text{launch}} + T_{\text{LFF}} + T_c > T_{\text{capture}} + T_{\text{hold}}$$



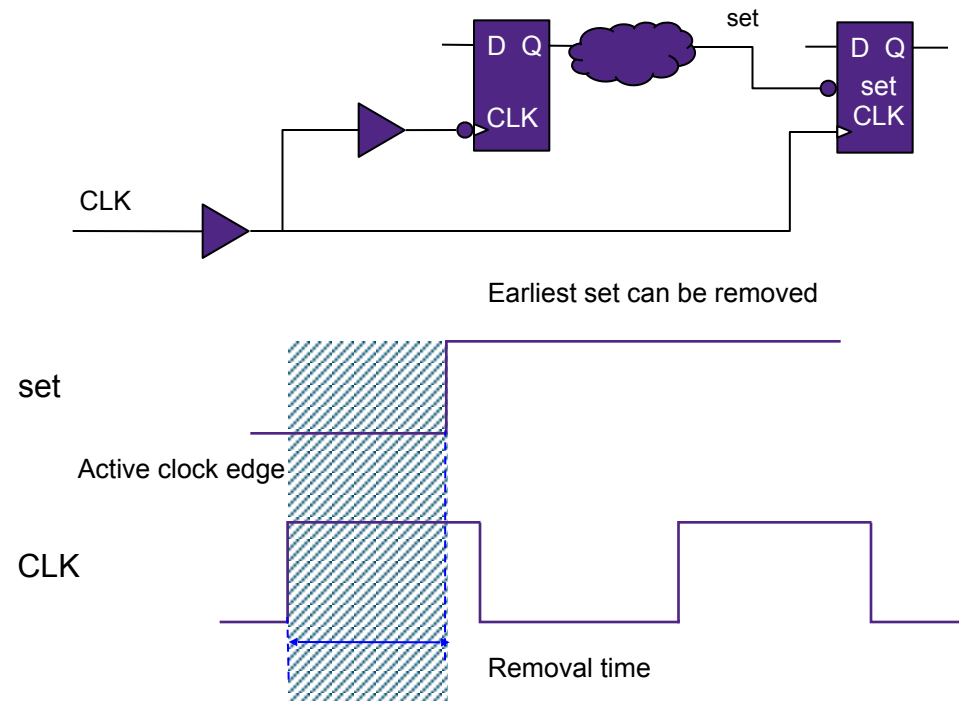
Hold Checks and Setup Check Cycles

- A hold timing check ensures that:
 - Data from the subsequent launch edge must not be captured by the setup receiving edge
 - Data from the setup launch edge must not be captured by the preceding receiving edge



Removal Timing Check

- Verifies that there is required amount of time between an active clock edge and the release of an asynchronous control signal



Min Pulse Width Checks

The minimum pulse width refers to the shortest duration of a pulse (high or low state) that a clock signal must maintain to be reliably recognized and processed by the digital circuit.

Clock pulse width violations can cause the following problems:

- Sequential devices (flip-flops and latches) might not capture data properly.
- In some logic circuits, the entire pulse could disappear

Typically, the pulse width shrinks because of the difference in the rise and fall delays of the cells. These differences arise because of variation effects which are more dominant in lower tech nodes, i.e.:

- Delay variation due to device aging. Rise and fall delays change at different rates, which causes min_pulse_width violation
- parts of clock tree age differently because of clock gating which causes increased clock skew and reduced min pulse width.

Min Pulse Width Checks

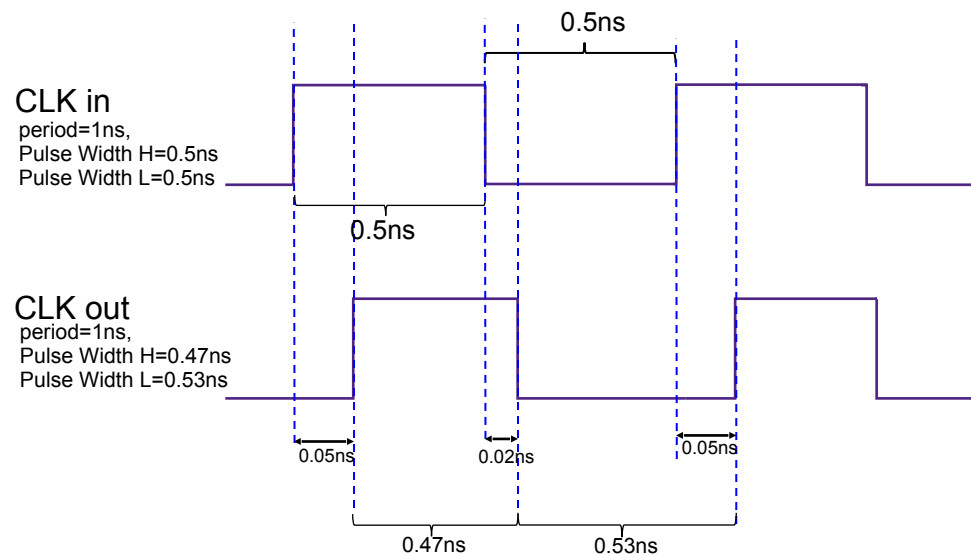
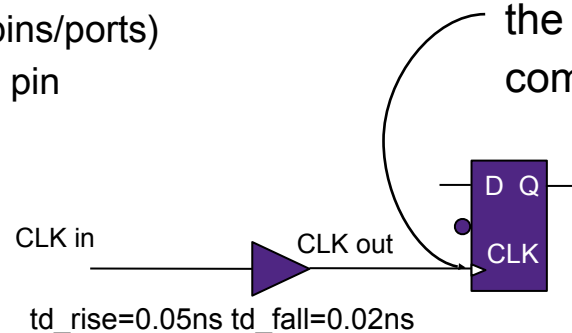
To set a min_pulse_width_check constraint:

- Use set_min_pulse_width command (on clocks/cells/ pins/ports)
- Or specify the requirement in .lib file for the respective pin

To report min pulse width use:

- report_min_pulse_width
- report_constraint -min_pulse_width

Min_pulse_width of CLK pin is usually included in the .lib. If not specify with set_min_pulse_width command



In this example if min_pulse_width constraint on flipflop/CLK pin is i.e. 0.5ns (for both high and low) The actual pulse width for high (0.47ns) will be violating the min pulse width constraint.

Thank You